

How English is like Kiitharaka and Búlí: successive cyclicity and optional wh-complementizer agreement in English

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1 Introduction

A growing consensus in syntax is that non-local dependencies can be analyzed as a *series of local dependencies*. One implementation of this idea is that long-distance dependencies are formed via *successive cyclic movement* (Chomsky, 1977) (see van Urk 2020 for a recent overview and Bouma et al. 2001 for a non-movement-based alternative using successive applications of feature percolation). Under this account, *which puppy* in (1) originates in the object position of *pet* and moves to the edge of the most deeply embedded clause. It then moves to the edge of each successive clause, eventually reaching the matrix clause, where it is pronounced.¹

- (1) [_{CP} Which puppy does Toma believe [_{CP} ~~which puppy~~ Mei said [_{CP} ~~which puppy~~ she pet ~~which puppy~~]]]?

The idea that long-distance dependencies are decomposed into a series of local dependencies explains a surprisingly wide range of data across many languages, including Belfast English (Henry, 1995), Búlí (Hiraiwa, 2005), Chamorro (Chung, 1998), Dinka (van Urk and Richards, 2015), Duala (Epeé, 1976), Ewe (Georgi, 2017; Collins, 1993), German (Müller, 1999), Haitian Creole (Takahashi and Gračanin-Yuksek, 2008), Hausa (Tuller, 1986), Indonesian/Malay (Saddy, 1991), Irish (McCloskey, 1979), Kiitharaka (Abels and Muriungi, 2008; Muriungi, 2005), Kikuyu (Clements, 1984), Moore (Haik, 1985), Spanish (Torrego, 1984), and Wolof (see van Urk 2020, Georgi 2017).

One salient piece of evidence for successive cyclic movement is that, in languages like Chamorro, Irish, and Wolof, the form of the complementizer (C) changes when it is in the *hierarchical path* of a long-distance dependency, that is, when the moving wh-element passes through the specifiers of CPs headed by the relevant Cs. For example, in Irish, C can be realized as *go* or *aL* (where L represents lenition of the following

¹For simplicity, we do not consider successive cyclic movement to the edge of vP here.

verb), depending on whether $\bar{A}$ -movement dependency (examples from McCloskey, 2002, cited in van Urk 2020). To aid comprehension, the relevant Cs are boxed.

- (2) a. Creidim gu-r inis sé bréag.
believe.1SG C.DECL-PAST tell he lie
‘I believe that he told a lie.’
b. an fhilíocht a chum sí.
the poetry C.EXT composed she
‘The poem that she composed.’
c. an t-ainm a hinnseadh dúinn a bhí ar an áit
the name C.EXT was.told to.us C.EXT was on the place
‘The name that we were told was on the place.’

In these examples, *go* (realized here as *gu*) occurs in a sentence with no crossing dependency, as in (2-a), and *aL* cannot replace it. By contrast, *aL* occurs when it is in the hierarchical path of a $\bar{A}$ -movement dependency, in this case one formed by relativization, as in (2-b) and (2-c). The C-form change triggered by long-distance dependencies is referred to as *wh-complementizer agreement* (see, e.g., van Koppen 2017; henceforth wh-C agreement). Phenomena that can be analyzed as wh-C agreement are also observed in other languages, including Búli, Chamorro, and Wolof. Under the standard assumption that wh-C agreement must be local, these phenomena suggest that the wh-phrase occupies the specifier position of the CP headed by the relevant C when agreement is computed. In other words, wh-phrases move successive cyclically, as in (1).

In this article, we report seven pairs of judgment and production experiments, with three major findings.

- English speakers were less likely to produce the overt C (*that*) when it is in the hierarchical path of an $\bar{A}$ -dependency, including those in wh-questions, relative clauses, comparatives, and topicalization. We call this effect the *that*-reduction effect.
- The acceptability-judgment analogue of this *that*-reduction effect was not reliably detectable.
- The *that*-reduction effect was selectively absent when the relevant C is the highest C inside the second conjunct of *Across-the-Board* (ATB) extraction sentences, as predicted by a proposal independently developed to explain reflexes of successive cyclic movement in unrelated languages (Georgi, 2019).

Based on these findings, we make four main arguments:

- English has *optional* wh-C agreement.

- ATB sentences involve *asymmetric* extraction from the first conjunct (Georgi, 2019).
- Even completely silent $\bar{A}$ -dependencies have a causal role in sentence formulation processes in production.
- Production data may provide evidence for syntactic mechanisms in ways that acceptability judgment data do not, or do so less clearly.

2 Previous Observations

The starting point of the current study is the observation that the optional C *that*, although clearly acceptable, is nevertheless only infrequently produced in wh-questions with object extraction from inside a complement clause (e.g., *Who did Mei believe (that) Toma met?*). For example, Phillips (2013) reports that in child-directed speech corpora consisting of 675,000 words (Pearl and Sprouse, 2013), *that* was used only twice in 161 instances of the relevant sentence type. That is, only around 1% of sentences of this type contained *that*. This is strikingly low, given that the use of *that* is fully acceptable in this context (see, e.g., Kim and Goodall 2024 and the acceptability data reported below; though see Cowart 1997; Tollan and Palaz 2021; Sobin 1987, 2002 for evidence that *that* may get slightly lower acceptability under some circumstances) and is commonly produced in comparable embedded clauses without a crossing wh-dependency.

The basic *that*-reduction effect observed in corpora has also been found in experimental settings with tighter controls. For example, Momma (2021) reports that speakers produced *that* in object wh-questions with cross-clausal extraction, like *Who does the artist think (that) the chef is chasing?*, less than 10% of the time across multiple experiments. Importantly, they also show that the same participants, performing the same task, produced *that* roughly half the time when producing sentences without a crossing wh-dependency (e.g., *Who thought (that) the chef is chasing the ballerina?*). This suggests that the *that*-reduction effect is not an artifact of the task and does not reflect a general avoidance of *that*. They also show that this effect occurs both in wh-questions and relative clauses, as in *the ballerina that the artist thinks (that) the chef is chasing is red.*, suggesting that, more generally, $\bar{A}$ -dependencies trigger the effect.

These previous studies suggest that *that* is less likely to be produced when it is in the hierarchical path of a $\bar{A}$ -dependency. However, to our knowledge, there is currently no explanation for this *that*-reduction effect.

3 The Current Hypothesis

Why do speakers use *that* only infrequently when it is in the hierarchical path of a $\bar{A}$ -dependency? As discussed above, Chamorro, Irish, and Wolof show that the form of a C is affected when it is in the hierarchical path of a $\bar{A}$ -dependency. The *that*-reduction effect fits this abstract description. Thus, we hypothesize that the *that*-reduction effect is a form of wh-C agreement. Under this analysis, speakers use the null C when it agrees with an operator in its specifier position, and this agreement is optionally realized morphologically, resulting in a null C. Optional realization of agreement has been typologically attested. For example, Wolof wh-C agreement is also optional, except in the terminal clause where the wh-phrase surfaces (Torrence, 2013, 2012). See also Levin et al. (2021) for optional verbal agreement in Santiago Tz’utujil. Also, while many reflexes of successive-cyclic movement add morphology or trigger syntactic effects such as inversion, there are also claims that some reflexes can lead to morphological reduction and even the absence of a morpheme (e.g., meN-deletion in Indonesian; Saddy 1991). Our hypothesis is that in English the reflex is realized as morphological reduction.

Under this *optional wh-C agreement* account of the *that*-reduction effect, we expect the effect to show characteristic properties of wh-C agreement in other languages, and models of wh-C agreement developed for those languages should be straightforwardly applicable to capturing the distribution of the *that*-reduction effect in English. The main goal of the current study is to test this general prediction.

4 Sentence Recall Task

The *that*-reduction effect is primarily a production phenomenon. As we will see below, an analogous effect in acceptability judgment, that is, the degradation of acceptability due to the presence of *that* in the hierarchical path of a $\bar{A}$ -dependency, is at best inconsistently observed. Studying it thus requires a method for reliably eliciting syntactically complex sentences with multiple clause embeddings and $\bar{A}$ -dependencies across clause boundaries. In this context, corpus analysis is limited by data sparsity and potentially innumerable confounds (see Roland et al. 2006 for various factors affecting C choice), and widely used elicitation methods, such as picture-description tasks, are limited in the range of sentences they can elicit.

Thus, here we use sentence recall tasks. In sentence recall tasks, participants are presented with target sentences, perform a distractor task, and then reproduce the sentences. A working assumption, widely adopted in the production literature as the *regeneration hypothesis* (Potter and Lombardi, 1990), is that sentence recall relies not on memory of surface form but on conceptual understanding and recently activated lexical items. Under this assumption, sentence recall involves production mechanisms, and hence the recalled sentences reflect speakers’ biases. Past studies have shown that production effects observed in naturalistic corpora can also be observed in sentence

recall studies, such as syntactic priming (Potter and Lombardi, 1998; Ferreira, 2003) and accessibility effects on word order (McDonald et al., 1993). Thus, the *that*-reduction effect could plausibly be observed in recalled sentences.

5 Experiment 1: Complement vs. relative clauses

Experiment 1 had two goals. The first was to establish that the *that*-reduction effect observed in previous studies can be replicated using a sentence recall task. The second was to evaluate whether the *that*-reduction effect can be explained by functional factors. Specifically, we tested whether the effect arises because speakers aim to minimize the linear distance between a wh-filler and the corresponding gap. In psycholinguistics, it is commonly assumed that longer dependencies are harder to process or produce, so speakers may prefer shorter dependencies between wh-phrases and their associated gaps (e.g., Futrell et al. 2015; Hawkins 1994, among others). Under this account, the *that*-reduction effect would arise because omitting *that* reduces dependency length by one word. We refer to this as the *linear distance minimization* account.

To evaluate the linear distance minimization account against our hypothesis that the *that*-reduction effect reflects optionally realized wh-C agreement, Experiment 1 compared sentences like the following:

- (3) a. Who did the cashier know (that) the robber attacked?
- b. Who did the robber (that) the cashier knew attacked?

In (3-a), the relevant C is in the hierarchical path of a wh-dependency. In (3-b), the relative clause C intervenes linearly, but not hierarchically, in the wh-dependency². The linear distance minimization account predicts that the *that*-reduction effect should be observed in both sentence types. The wh-agreement account predicts that the effect should be observed selectively in sentences like (3-a).

5.1 Methods

5.1.1 Participants

For the acceptability judgment task, 55 self-reported monolingual English speakers were recruited via Prolific Academic (18 residing in the United Kingdom, 17 in the United States, 2 in Canada, and the rest unreported; mean age = 42, range = 26–66). For the production (typing) task, 104 self-reported monolingual English speakers were recruited (27 residing in the United Kingdom, 42 in the United States, 2 in Canada, 1 in Australia, and the rest unreported; mean age = 43, range = 24–76). Participants

²*that* in (3-b) hierarchically intervenes in the operator–variable dependency created by relativization. However, this dependency likely terminates in the specifier of *that* and therefore, under the account developed later in this paper, does not trigger the *that*-reduction effect.

were compensated at a rate of \$10/hour. The acceptability task took approximately 15 minutes, and the production task took approximately 30 minutes.

5.1.2 Materials and design

The production task used a 2 (Clause Type: Relative Clause (RC) vs. Complement Clause (CC)) $\times$ 2 (Complementizer: *that* vs. null) $\times$ 2 (Dependency Type: Crossing vs. Local) within-subject and within-item design. For critical items, 24 sentence sets were constructed, such as:

- (4) a. Who did the robber (that) the cashier knew attack? - A sheriff.
- b. Who was attacked by the cashier (that) the robber knew? - A sheriff.
- c. Who did the robber believe (that) the cashier knew? - A sheriff.
- d. Who believed (that) the cashier knew the robber? - A sheriff.

In (4-a), the C is inside the relative clause modifying the subject (*the robber*), so it only linearly intervenes in the wh-dependency. (4-b) serves as the control, where the wh-dependency is resolved locally in the matrix subject position. In (4-c), the C is inside the complement clause from which the wh-phrase is extracted, so it hierarchically (and linearly) intervenes in the wh-dependency. (4-d) serves as the control for CC, where the wh-dependency is resolved locally. Each sentence type had a version with *that* and a version without it. Thus, there were a total of 8 conditions.

Critical items were distributed using a Latin square design to create eight lists. Each list was divided into eight blocks. Each block contained four question-answer pairs, one from each condition. The four pairs within a block were randomly ordered for each participant, although the order of blocks was kept constant across participants. This means that members of the same item set appeared at approximately the same position across lists.

For the acceptability task, only sentences with crossing dependencies were tested because (a) sentences with local dependencies were uncontroversially acceptable, and (b) our primary interest was whether the presence of *that* lowers acceptability when the wh-dependency crosses the complementizer. Thus, there were four conditions: RC crossing with *that*, RC crossing without *that*, CC crossing with *that*, and CC crossing without *that*. Sentences were distributed across four lists and presented in random order. Filler sentences of four types were included, each containing at least a mild violation: (1) extraction of a *which*-phrase from a *whether*-clause (whether-island (which)), (2) extraction of a *how many*-phrase from a *whether*-clause (whether-island (how)), (3) *how*-questions with negation (How + negative), and (4) simultaneous violations of wh-islands and subcategorization constraints (multiple violation). Each filler type contained six sentences, for a total of 24 fillers. Fillers were intermixed with critical sentences.

All materials are available on OSF (<https://tinyurl.com/376dvfnd>).

5.1.3 Procedure

Each block began with four question–answer pairs presented one at a time. Participants were instructed to read and memorize each pair. Each pair remained on the screen until the space key was pressed. After memorization, answers (without questions) were presented in the same order. Participants were instructed to recall the corresponding question and type it into a textbox. Each answer remained on the screen until the space key was pressed.

5.1.4 Scoring

All typed responses were coded by the first author. Responses were coded as not analyzable if the target structure was not preserved or if they did not form a complete sentence. Responses were coded as analyzable even if words were replaced, as long as the target structure was preserved. Sentences containing typos were coded as analyzable if the intended sentence was clear.

5.1.5 Analysis

All analyses were conducted using R (R Core Team, 2021) and the `brms` package (Bürkner, 2021). For each Bayesian hierarchical model, four Markov chain Monte Carlo chains with 5,000 iterations each were run, discarding the first 2,000 iterations as warm-up. All $\hat{R}$ values were 1.00. Effects were considered present when the 95% credible interval of the slope estimate did not include 0.

For the acceptability data, acceptability scores were z-scored within each subject and analyzed using Bayesian hierarchical mixed-effects regression with Clause Type (RC vs. CC; sum-coded -0.5, 0.5), Target Type (*that* vs. null; sum-coded -0.5, 0.5), and their interaction as fixed effects. The maximal random-effects structure was included (Barr et al., 2013). Priors were: normal(0, 1.5) for the intercept, normal(0, 0.5) for slopes, exponential(1) for random-effect standard deviations, and LKJ(2) for correlation matrices.

For the production data, coded responses were analyzed using Bayesian hierarchical mixed-effects regression, assuming a Bernoulli distribution with a logit link function. Fixed effects included Dependency Type (Crossing vs. Local; sum-coded -0.5, 0.5), Clause Type (RC vs. CC; sum-coded -0.5, 0.5), Target Type (*that* vs. null; sum-coded -0.5, 0.5), and their interactions. The maximal random-effects structure was included (Barr et al., 2013). Priors were: normal(0, 1.5) for the intercept, normal(0, 1) for slopes, exponential(1) for random-effect standard deviations, and LKJ(2) for correlation matrices.

5.2 Results and discussion

5.2.1 Acceptability

Figure 1 shows mean ratings across conditions and for filler sentences. There was no strong evidence that participants rated CC sentences with *that* differently from CC sentences without ($\hat{\beta} = -0.09$, 95% CrI = $[-0.25, 0.06]$). RC sentences with *that* were rated more acceptable than those without ($\hat{\beta} = -0.31$, 95% CrI = $[-0.47, -0.15]$), possibly due to the *filled-gap effect* (Stowe, 1986). When an embedding verb has an NP-taking bias (e.g., *disclose* in *Who did the committee disclose the dean consulted?*), comprehenders may temporarily misanalyze *the dean* as the direct object of *disclose*, causing comprehension difficulty. With *that* after the verb, this temporary ambiguity is avoided, explaining lower ratings for sentences without *that*.

There was evidence for a main effect of Complementizer ($\hat{\beta} = 0.20$, 95% CrI $[0.07, 0.32]$), qualified by a Complementizer $\times$ Clause Type interaction ($\hat{\beta} = 0.22$, 95% CrI $[0.03, 0.41]$). This suggests that *that* improved ratings for RC sentences but not CC sentences. RC sentences were overall rated lower (main effect of Clause Type: $\hat{\beta} = -0.89$, 95% CrI $[-1.09, -0.78]$), as expected given doubly center-embedded structures.

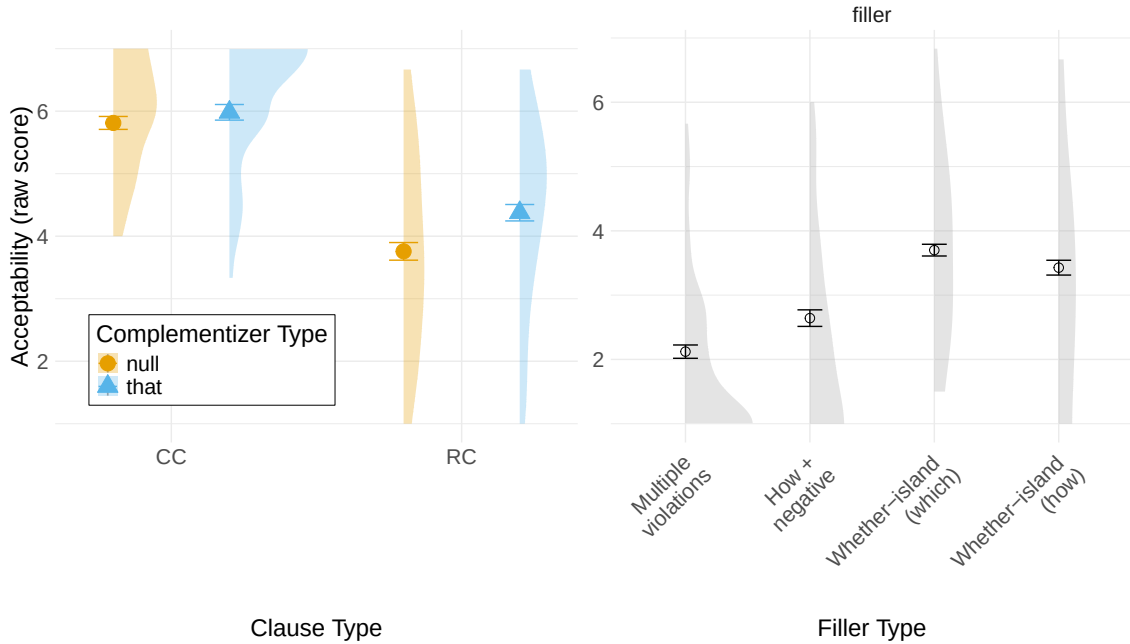


Figure 1: Mean ratings for critical (left) and filler (right) items in Experiment 1. Red and blue regions represent subject averages. Error bars are within-subject standard errors.

5.2.2 *That*-production

1,098 of 2,732 trials (40.0%) met the scoring criteria (CC crossing: 42.9%, CC local: 55.9%, RC crossing: 25.2%, RC local: 36.7%).

Figure 2 shows the mean rate of *that* production across conditions. Speakers were less likely to produce *that* in CC sentences with a crossing wh-dependency ($\hat{\beta} = -1.05$, 95% CrI = $[-1.64, -0.52]$). No analogous effect was found in RC sentences ($\hat{\beta} = 0.02$, 95% CrI = $[-0.69, 0.69]$). This contrast was supported by a Clause Type $\times$ Dependency Type interaction ($\hat{\beta} = -1.07$, 95% CrI = $[-1.95, -0.22]$). There was also a main effect of Target Type ($\hat{\beta} = 1.41$, 95% CrI = $[0.79, 2.07]$), reflecting participants’ memory for the original sentences.

Taken together, these results suggest that (i) the *that*-reduction effect can be observed in recall-based production studies, (ii) the analogous effect in acceptability judgment is weak or absent, and (iii) dependency-length minimization is unlikely to explain the *that*-reduction effect.

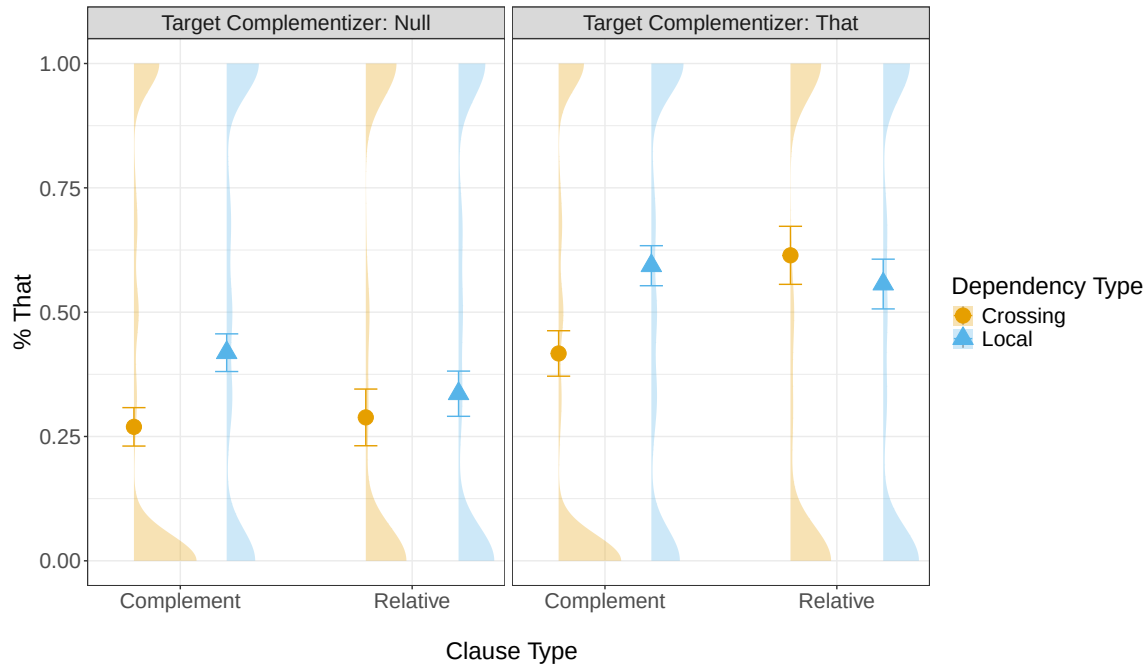


Figure 2: Mean rate of *that* use across conditions in Experiment 1. Error bars are within-subject standard errors.

6 Experiment 2: Argument vs. adjunct wh

Experiment 1 suggests that the *that*-reduction effect in production is unlikely to result from speakers or typists attempting to minimize the linear distance between the

filler and the gap. However, the effect could still be explained in functional terms. For example, the *that*-reduction effect could arise from differences in the amount of material following the potential *that* site. In *Who did the robber believe (that) the cashier knew?*, the material after *that* is simply *the cashier knew*. In contrast, in *Who believed (that) the cashier knew the robber?*, the material after *that* is the full embedded clause *the cashier knew the robber*. Psycholinguistic research suggests that optional elements like *that* are more likely to be produced when upcoming material is harder to plan (Roland et al., 2006; Ferreira, 2003). Thus, the *that*-reduction effect could arise simply because speakers produce *that* more often when more material follows it. This account, with some auxiliary assumptions, may also explain the absence of a *that*-reduction effect in the RC sentences in Experiment 1. In the RC conditions, *that* precedes *the cashier knew attack?* in the crossing dependency condition and *the cashier knew the robber?* in the local dependency condition. Thus, speakers have more words to plan after *that* in CC sentences with crossing than local dependencies, but not in RC sentences. This may have produced the selective *that*-reduction effect in CC sentences observed in Experiment 1.

Experiment 2 assessed the plausibility of this planning-based account by investigating adjunct wh-questions, which do not differ in the number of words preceding or following *that*, regardless of whether the dependency is locally resolved or crosses a clause boundary. It also tested whether the *that*-reduction effect generalizes to adjunct wh-questions.

6.1 Methods

6.1.1 Participants

For the forced-choice judgment task, 42 speakers of English as their first language were recruited via Prolific Academic (11 residing in the United Kingdom, 10 in the United States, 2 in Australia, 2 in Canada, and the rest did not provide country information; mean age = 33, age range = [21, 73]).

For the production task, 40 participants were recruited from the local community at the University of Massachusetts, Amherst. Four participants were replaced due to recording failures.

6.1.2 Materials

Twenty-four question–answer pair sets like the following were created:

- (5) Where did the biologist say (that) a new species was discovered?
 - a. At a conference. [Matrix scope]
 - b. On an isolated island. [Embedded scope]

In the adjunct extraction conditions, the target questions that participants memorized

were constant across conditions, but the answers were manipulated to create the matrix scope condition as in (5-a) and the embedded scope condition as in (5-b).

Twenty-four additional sets of question–answer pairs were created for argument extraction conditions:

- (6) a. Who said (that) the pirate attacked the sailor? [Matrix scope]
- b. Who did the sailor say (that) the pirate attacked? [Embedded scope]
- c. A merchant.

Here, the answer was constant across matrix and embedded extraction conditions, but the question differed. In (6-a), the *wh*-phrase is associated with the matrix clause, and in (6-b), it is associated with the embedded clause.

For the production task, the 48 sets of question–answer pairs were distributed across eight experimental lists using a Latin square design. Each list was divided into 12 mini-blocks, each containing four question–answer pairs. Within each mini-block, the four pairs never belonged to the same condition. Specifically, exactly two target questions contained *that*, exactly two had matrix *wh*-scope, and exactly two were adjunct *wh*-questions.

For the forced-choice task, only the adjunct questions were used as critical stimuli. The 24 adjunct questions (paired with both embedded-clause and matrix-clause answers) were intermixed with 24 filler questions (e.g., *Who said the pirate attacked the sailor?*), each paired with both plausible (e.g., *A merchant*) and implausible answers (e.g., *A dog*).

In addition, we included a further condition (the *whether* condition) in which *that* in the *that* condition was replaced with *whether*, as in *Where did the biologist say whether a new species was discovered?*, to assess whether the embedded scope is indeed blocked in the context of islands.

6.1.3 Procedure

Experiment 2 followed mostly the same procedure as Experiment 1, with two notable differences.

The first difference is that we used a forced-choice task instead of an acceptability judgment task. Participants indicated whether they interpreted each question as referring to the matrix or embedded clause. The embedded-clause interpretation (where the *wh*-dependency crosses the complementizer) and the matrix-clause interpretation (where the dependency is resolved locally) were string-identical. Participants read a question (e.g., *Where did the biologist say a new species was discovered?*) and chose between two alternative answers corresponding to either the embedded-scope interpretation (e.g., *On an isolated island*) or the matrix-scope interpretation (e.g., *At a conference*). The dependent variable was the chosen answer, serving as a proxy for participants’ interpretation of the scope-ambiguous questions.

The second difference is that we elicited spoken responses instead of typed responses. Otherwise, the task structure was identical to Experiment 1.

6.1.4 Scoring and analysis

The same scoring and analysis procedures as in Experiment 1 were used.

6.2 Results and discussion

6.2.1 Forced choice

The results are shown in Figure 3. In filler trials where one of the choices was highly implausible, participants chose the plausible answer 88% of the time, suggesting that they were completing the task as intended. In critical trials, participants chose the embedded-scope answer 57.0%, 51.4%, and 31.8% of the time in the null, *that*, and *whether* conditions, respectively. Thus, participants were around 5.6% more likely to choose the matrix-scope answer when *that* was present compared to when it was absent. However, there was at best weak statistical evidence for this effect ($\hat{\beta} = -0.32$, $SE = 0.21$, 95% CrI = [-0.73, 0.10]). In contrast, participants were 25.0% more likely to choose the matrix-scope answer in the *whether* condition than in the other two conditions, with strong evidence for this effect ($\hat{\beta} = -1.25$, $SE = 0.18$, 95% CrI = [-1.61, -0.90]).

This effect suggests that embedded-scope answers were indeed associated with embedded-clause scope interpretations, which are blocked, though not categorically, when *whether* intervenes in the wh-dependency.

Thus, the forced-choice results provide no clear evidence that the matrix interpretation is more likely with *that* than without it, suggesting that the embedded interpretation is compatible with *that*.

6.2.2 That-production

871 of 1,427 (61%) trials met the scoring criteria. The proportion of analyzable responses was 62.1% for embedded-scope adjunct-wh sentences with *that*, 68.9% for embedded-scope adjunct-wh sentences without *that*, 69.1% for matrix-scope adjunct-wh sentences with *that*, 68.4% for matrix-scope adjunct-wh sentences without *that*, 45.9% for embedded-scope argument-wh sentences with *that*, 46.6% for embedded-scope argument-wh sentences without *that*, 64.4% for matrix-scope argument-wh sentences with *that*, and 63.2% for matrix-scope argument-wh sentences without *that*.

The results are shown in Figure 4. For both argument and adjunct extractions, speakers were less likely to use *that* when wh-phrases scoped over embedded clauses than over matrix clauses. There was clear evidence for a main effect of Scope ($\hat{\beta} = 1.47$, $SE = 0.30$, 95% CrI = [0.90, 2.08]), with no evidence for a Scope $\times$ Extraction Type interaction ($\hat{\beta} = 0.36$, $SE = 0.44$, 95% CrI = [-0.50, 1.24]). There was only

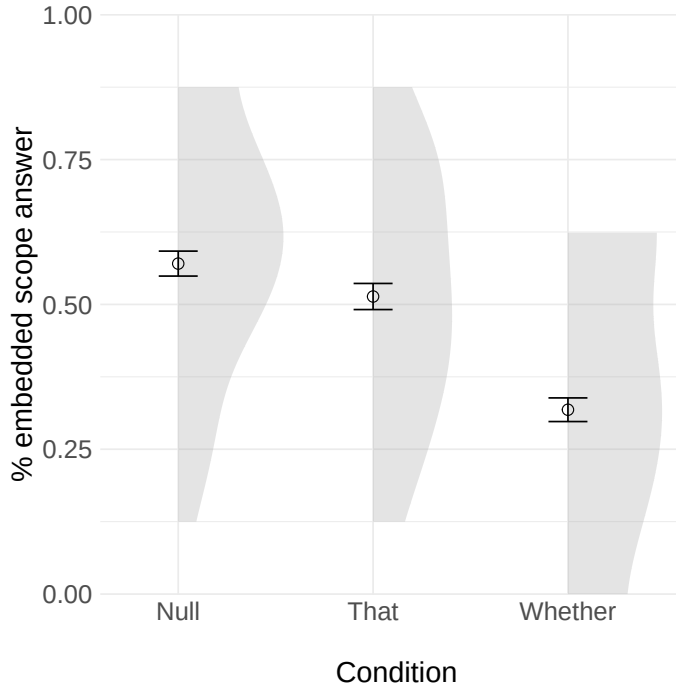


Figure 3: Mean proportion of embedded scope answer choice in Experiment 2. Error bars are within-subject standard errors.

weak evidence for a Target Type effect; speakers were slightly more likely to produce *that* when the original sentence contained it ($\hat{\beta} = 0.50$, $SE = 0.29$, 95% CrI = [-0.07, 1.06]). No other effects were observed.

Overall, the production results revealed clear evidence for the *that*-reduction effect, while the forced-choice task revealed weak or no evidence for an analogous effect in scope judgment. In Experiment 2, the linear position of *that* was held constant across embedded- and matrix-scope adjunct wh-questions. Thus, the *that*-reduction effect generalizes to adjunct wh-extraction, and it is unlikely to be driven by a simple strategy of minimizing the linear distance between the filler and the gap.

7 Experiment 3: Across three clauses

As illustrated by the Irish examples, wh-C agreement is often realized on all complementizers along the path of $\bar{A}$ -dependencies. To test the parallel between wh-C agreement and the *that*-reduction effect, Experiment 3 examined whether the *that*-reduction effect extends to all complementizer positions along the hierarchical path of a wh-dependency in multiply embedded structures.

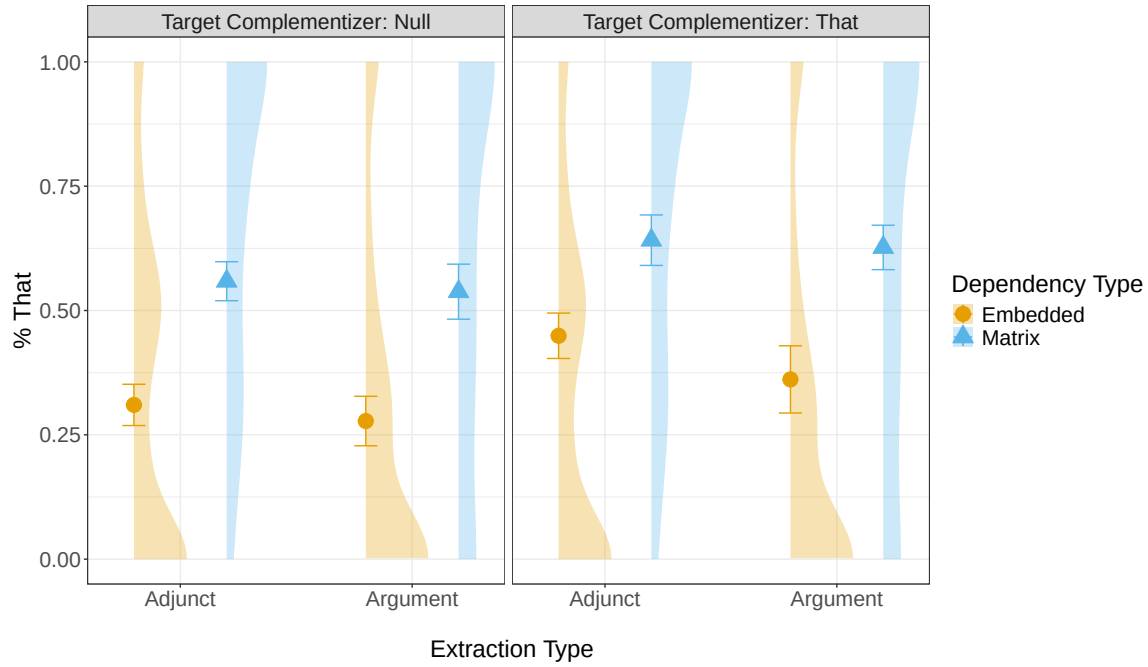


Figure 4: Mean proportion of *that* in each condition in Experiment 2. Error bars are within-subject standard errors.

7.1 Methods

7.1.1 Participants

For the acceptability judgment task, 24 self-reported English monolinguals were recruited via Prolific Academic (16 residing in the U.S., 8 residing in the U.K.; mean age = 45, age range = [24, 68])³. For the production task, 34 self-reported English monolinguals were recruited (25 residing in the U.S., 9 in the U.K.; mean age = 43, age range = [32, 71]).

7.1.2 Materials

Twenty-four sets of critical sentences like the following were constructed:

- (7) a. Which story did Jane imply (that) her son believed (that) she came up with? [crossing]
 b. Jane implied (that) her son believed (that) she came up with the story. [local]

³Here and in subsequent experiments, we did not have participants with unidentified country of residence, due to changes in filtering settings in Prolific. We recruited monolingual English speakers who resided in the U.S. or the U.K.

Sentence (7-a) involves two complementizers in the path of a wh-dependency. Sentence (7-b) is the control sentence with no wh-dependency. In Experiment 3, we used a D-linked wh-phrase so that the wh-questions and control sentences were maximally similar in lexical content.

Each critical sentence was paired with a randomly generated declarative sentence with no obvious relationship to the critical sentence (e.g., *A mechanic fixed an engine to restore the vehicle.*). These sentences served as distractors designed to prevent participants from rehearsing the target sentence.

Six filler sentence pairs were constructed and intermixed with the critical sentence pairs. These filler pairs were structurally similar to the critical pairs. In filler trials, participants were given lexical cues corresponding to the second sentence, rather than the first. Thus, in critical trials participants recalled the first sentence, whereas in filler trials they recalled the second sentence. This was intended to prevent participants from simply memorizing the first sentence without attending to the second sentence.

See the OSF repository for the full set of stimuli and the lexical cues used for each sentence.

7.1.3 Procedure

In Experiment 3 and all subsequent experiments, we used a version of the sentence recall task that differed from those used in Experiments 1 and 2. This is because memorizing more than one sentence at a time (the requirement of the task used in Experiments 1 and 2) would have been too difficult for the types of sentences investigated in Experiment 3 and subsequent experiments.

In Experiment 3 and all subsequent experiments, participants were instructed to read two sentences, one sentence at a time. They were given up to 7 seconds to read aloud and memorize each sentence. After that, they were presented with four sub-chunks of either the first sentence (in critical trials) or the second sentence (in filler trials), which served as memory cues for sentence recall. Each sub-chunk was presented in a corner of the screen (see Figure 5). Participants were given up to 15 seconds to produce the target sentence. The lexical cues corresponded to content words from the target sentence, plus *which* or *the* accompanying the head of the extracted noun phrase. For example, for the target sentence *Which story did Jane imply (that) her son believed (that) she came up with?*, we provided *Which story*, *Jane imply*, *son believed*, and *came up with?*. For the minimally different control sentence *Jane implied (that) her son believed (that) she came up with the story.*, we provided *Jane implied*, *son believed*, *came up with*, and *the story*.

We provided these sub-chunks as memory cues to ensure that participants could reconstruct the target sentence structures with the intended embedding verbs even if they partially forgot the original sentence (i.e., to minimize data exclusion). Ensuring that participants used the prescribed embedding verbs is also useful because it is well known that the rate of *that* is strongly influenced by the identity of embedding verbs

(Roland et al., 2006).

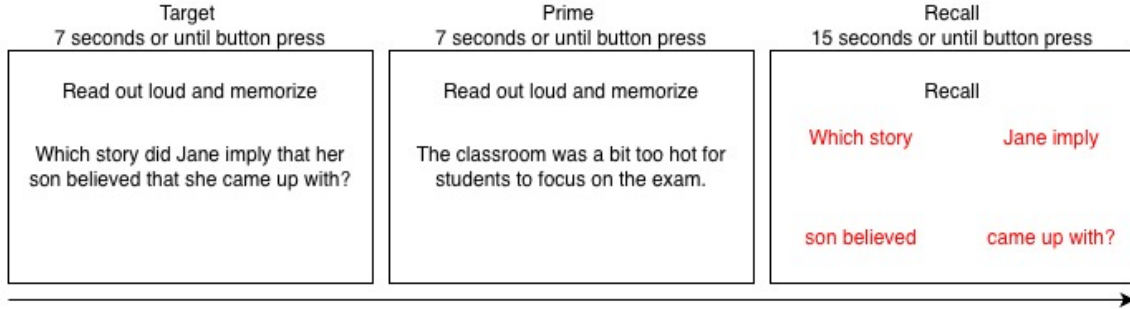


Figure 5: Schematic illustration of a single trial in the sentence recall task (used in Experiments 3-7).

7.1.4 Scoring and analysis

The same scoring procedure as in Experiments 1-2 was used, except that in Experiment 3 onward we also excluded responses in which the target sentence was not read aloud with the correct complementizer (e.g., when the target sentence contained *that* but participants omitted *that* during reading for memorization, the corresponding response was excluded). The analysis procedure for the acceptability judgment data was identical to that used in the previous experiments.

7.2 Results

7.2.1 Acceptability

As shown in Figure 6, all sentences were rated as moderately acceptable. This is consistent with the presence of two clause embeddings (e.g., *Which story did Jane imply (that) her son believed (that) she came up with?*), which naturally increases processing difficulty and can lower overall acceptability in experimental settings.

As expected, there was evidence that sentences with crossing dependencies were rated lower than control sentences with local dependencies (Dependency Type: $\hat{\beta} = 0.27$, 95% CrI = [0.13, 0.41]). More importantly, there was no evidence that the presence or absence of *that* in either the first or second complementizer position affected acceptability (First Complementizer: $\hat{\beta} = 0.06$, 95% CrI = [-0.18, 0.30]; Second Complementizer: $\hat{\beta} = -0.07$, 95% CrI = [-0.19, 0.05]). No other effects were observed.

7.2.2 That-production

760 of 816 (93.1%) trials met the scoring criteria. The proportion of analyzable responses was 94.1% for control sentences with *that* in neither clauses, 91.2% for wh

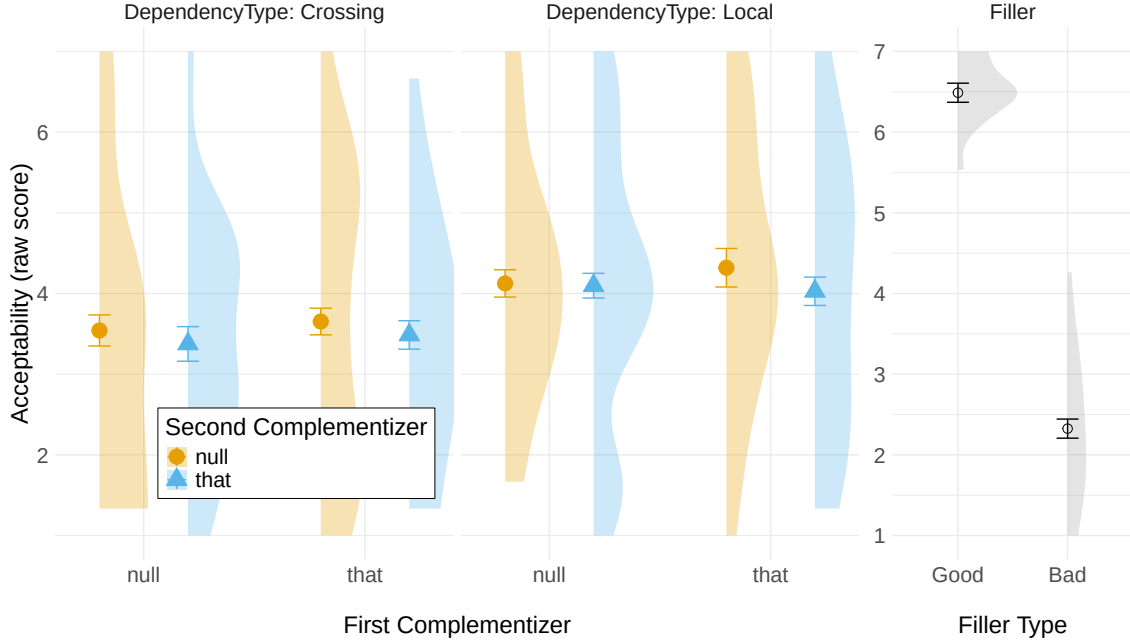


Figure 6: Mean rating of critical sentences used in Experiment 3. Error bars are within-subject standard errors.

questions with *that* in neither clauses, 94.1% for control sentences with *that* only in the second clause, 94.1% for wh questions with *that* only in the second clause, 93.1% for control sentences with *that* in the first clause only, 93.1% wh questions with *that* in the first clause only, 95.1% for control sentences with *that* in both clauses, and 90.2% for wh questions with *that* in both clauses.

As shown in Figure 7, in the first complementizer position participants were less likely to use *that* with a crossing wh-dependency than without ($\hat{\beta} = -1.16$, 95% CrI = [0.56, 1.72]). Likewise, in the second complementizer position, they were less likely to use *that* with a crossing wh-dependency than without ($\hat{\beta} = 1.86$, 95% CrI = [1.34, 2.45]). There was clear evidence for a main effect of Dependency Type ($\hat{\beta} = -1.52$, 95% CrI = [-1.91, -1.15]), with no clear evidence for an interaction between Dependency Type and Complementizer Position ($\hat{\beta} = -0.70$, 95% CrI = [-1.57, 0.13]). This suggests that the magnitude of the *that*-reduction effect did not differ reliably between the two complementizer positions.

There was also evidence for a main effect of Target Type ($\hat{\beta} = 1.49$, 95% CrI = [0.96, 2.05]), suggesting that speakers retained some memory of the original sentences. There was evidence for a main effect of Position ($\hat{\beta} = 1.19$, 95% CrI = [-1.91, -0.51]). No other effects were observed.

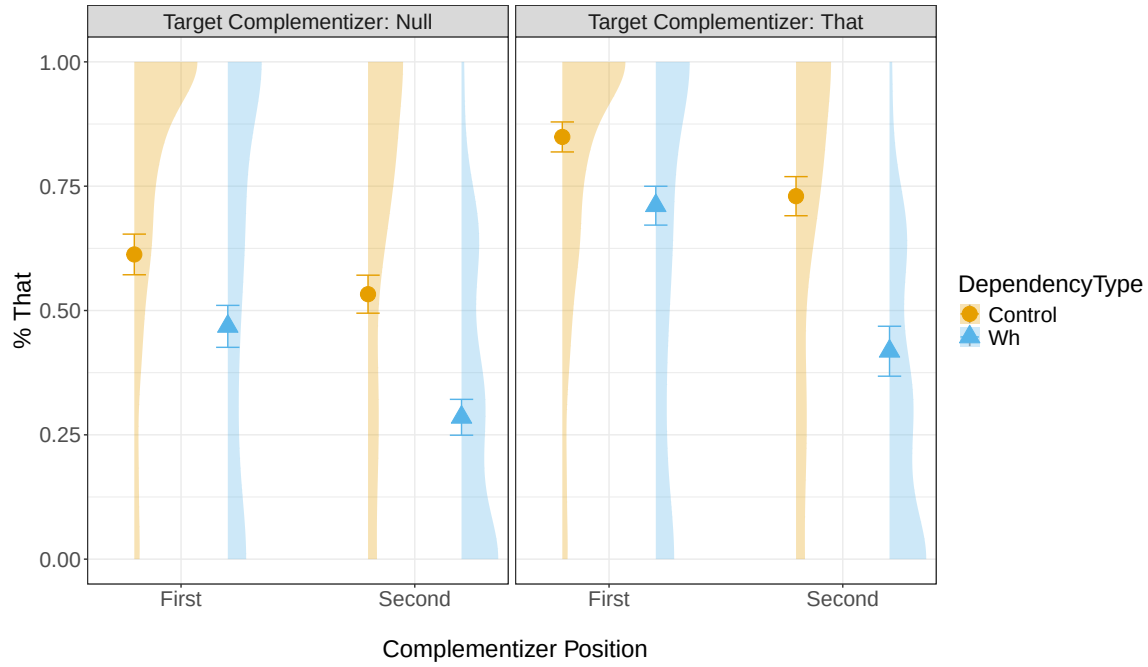


Figure 7: Mean proportion of *that* in each condition in Experiment 3. Error bars are within-subject standard errors.

8 Experiment 4: Comparatives

In Experiment 3, the *that*-reduction effect was observed when the relevant complementizer position was in the hierarchical path of a *wh*-dependency. Experiment 4 tested whether the *that*-reduction effect generalizes to other $\bar{A}$ dependencies. This parallels the observation that complementizer agreement in other languages is not limited to *wh*-dependencies, but also extends to other $\bar{A}$ dependencies such as relative clauses and comparatives (McCloskey, 2001).

There is already some evidence that the *that*-reduction effect generalizes to $\bar{A}$ dependencies introduced in relative clauses. Specifically, Momma (2021) reported a *that*-reduction effect in relative clauses such as *The ballerina that the artist thinks (that) the chef is chasing*. The main goal of Experiment 4 is to test whether the *that*-reduction effect extends further to other $\bar{A}$ dependencies, namely those involved in comparative constructions.

Comparatives are often analyzed as involving an $\bar{A}$ dependency (Chomsky, 1977), akin to that found in *wh*-questions. The following example illustrates this analysis, where *x-many* corresponds to the degree operator forming an $\bar{A}$ dependency.

- (8) Toma ate more cookies than ~~x-many cookies~~ Mei claimed ~~x-many cookies~~ he did eat ~~x-many cookies~~.

There is converging evidence for this analysis. First, the degree operator is overtly expressed as a *wh*-word in some varieties of American English, as noted by Chomsky (1977):

- (9) John is taller than what Mary is.

Second, comparatives are sensitive to island constraints (Lechner and Corver, 2017). For example:

- (10) *Toma ate more cookies than Mei had a plan to eat.

See Lechner and Corver (2017) for a more extensive list. To the extent that island effects diagnose $\bar{A}$ dependencies, these examples suggest that comparatives involve an $\bar{A}$ dependency inside the *than*-clause. Experiment 4 thus tested whether the *that*-reduction effect is observed inside the *than*-clause in comparatives.

8.1 Methods

8.1.1 Participants

For the acceptability judgment task, 24 self-reported monolingual speakers of English were recruited via Prolific Academic (8 residing in the U.S., 16 residing in the U.K.; mean age = 51, age range = [29, 77]). For the production study, 36 self-reported monolingual speakers of English were recruited (26 residing in the U.S., 10 residing in the U.K.; mean age = 41, age range = [20, 67]). Four participants were replaced because they did not follow the instructions.

8.1.2 Materials

Twenty-four critical sentence sets like the following were constructed.

- (11) a. The scientist wrote more papers than her co-worker assumed (that) she did.
b. The scientist wrote many papers, and her co-worker assumed (that) she did.

Sentence (11-a) is a comparative sentence with a complement clause inside the *than*-clause, optionally containing *that*. Sentence (11-b) is a minimally different control sentence with coordination instead. Control sentences were constructed by replacing *more* with *many* and *than* with *and* (with a preceding comma). For clause-embedding verbs (e.g., *assume*), we used 12 distinct verbs (e.g., *think*, *assume*, *claim*, *believe*), each used twice.

For memory cues, we provided content words from the target sentences, along with *more* and *than* (for comparative sentences) or *many* and *and* (for control sentences), as well as the subject of the final clause and the auxiliary licensing VP ellipsis (e.g.,

she did). For example, for (11-a), we provided *scientist wrote, more papers than, co-worker assumed*, and *she did* as memory cues.

As in Experiment 3, each critical sentence was paired with an unrelated declarative sentence that served as a distractor, intervening between the memorization and recall of the critical sentence. The same set of distractor sentences as in Experiment 3 was used.

In addition, four filler sentence pairs that were structurally similar to the critical pairs were also constructed and intermixed with the critical sentence pairs.

8.1.3 Procedure, scoring, and analysis

The procedure, scoring, and analysis methods were the same as in Experiment 3.

8.2 Results

8.2.1 Acceptability

Figure 8 shows mean acceptability ratings for each condition. Comparative sentences were rated lower than control sentences ($\hat{\beta} = -0.55$, 95% CrI = $[-0.78, -0.32]$), likely because they are pragmatically more complex to accommodate.

This effect was qualified by a Sentence Type $\times$ Complementizer interaction ($\hat{\beta} = 0.20$, 95% CrI = $[0.00, 0.39]$). There was some evidence that *that* slightly decreased acceptability in comparative sentences ($\hat{\beta} = 0.16$, 95% CrI = $[0.02, 0.30]$), but not in control sentences ($\hat{\beta} = -0.04$, 95% CrI = $[-0.18, 0.11]$). No evidence was found for other effects.

8.2.2 *That* production

820 of 860 trials (94.9%) met the scoring criteria (96.8% for comparative sentences with *that*, 98.6% for comparative sentences without *that*, 93.1% for control sentences with *that*, and 91.2% for control sentences without *that*).

Figure 9 shows the mean rate of *that* use across conditions. Speakers were less likely to use *that* inside *than*-clauses in comparative constructions than in control sentences ($\hat{\beta} = 0.61$, 95% CrI = $[0.19, 1.05]$). Speakers also used *that* more often when the original sentence contained *that* ($\hat{\beta} = 1.48$, 95% CrI = $[0.88, 2.10]$), suggesting some memory for the original target sentences. There was no evidence for the interaction between Sentence Type and Target Complementizer ($\hat{\beta} = -0.20$, 95% CrI = $[-1.24, 0.83]$).

The current results suggest that the *that*-reduction effect generalizes to a broad class of $\bar{A}$ dependencies, including at least those involved in wh-questions, relative clauses, and comparatives. In this experiment, there was also some evidence for an analogous effect in the acceptability data.

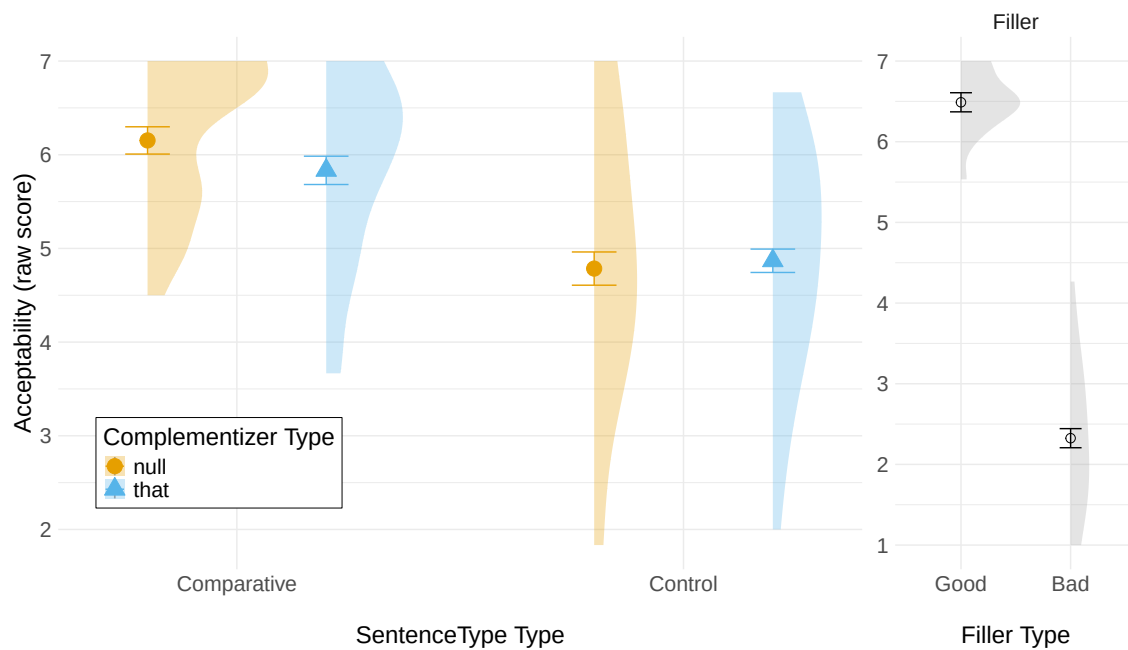


Figure 8: Mean rating of critical sentences used in Experiment 4. Error bars are within-subject standard errors.

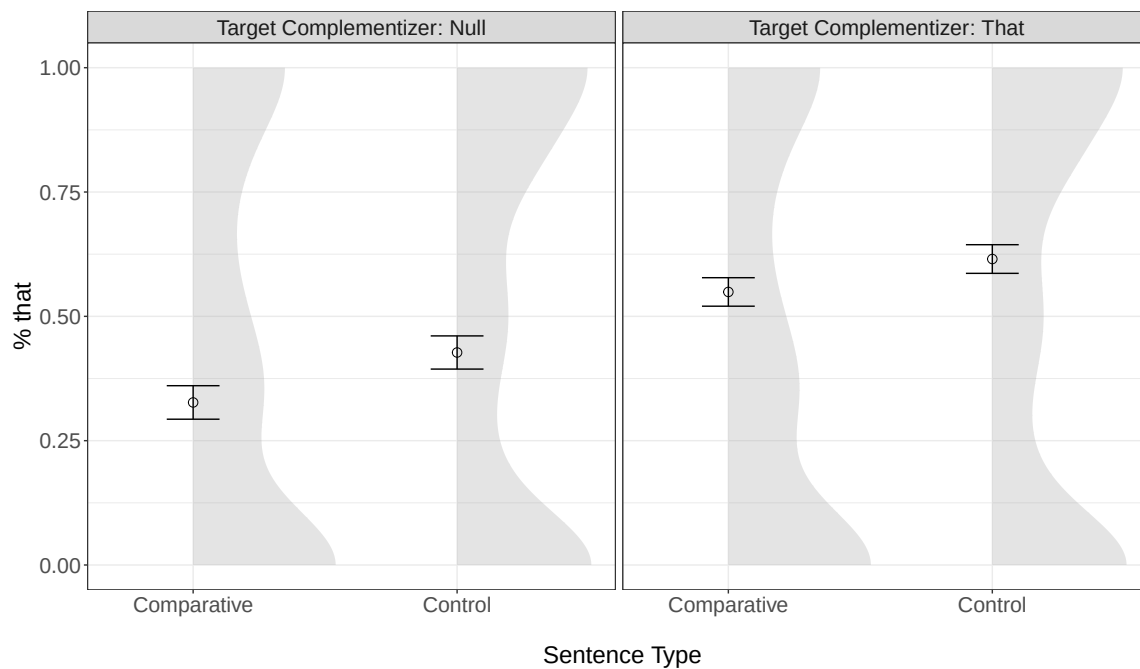


Figure 9: Mean proportion of *that* in each condition in Experiment 4. Error bars are within-subject standard errors.

Experiment 4 further suggests that speakers represent $\bar{A}$ dependencies even when neither the filler, the gap, nor the gap-hosting predicate is overtly pronounced. Under standard analyses of comparatives involving filler-gap dependencies, sentences such as *The scientist wrote more papers than her co-worker assumed (that) she did* are analyzed as follows:

- (12) The scientist wrote more papers than ~~x-many papers~~ her co-worker assumed (that) ~~x-many papers~~ she did write ~~x-many papers~~.

On such an analysis, the dependency is fully abstract in the sense that neither the degree operator (represented as x-many) nor the gap-hosting predicate is overtly realized. Moreover, because the gap is contained within an elided VP, this $\bar{A}$ dependency also implies that elided VPs have internal structure from which the degree operator can be extracted.

Some grammatical theories, including certain versions of Combinatory Categorical Grammar (Steedman, 2001), Construction Grammar (Goldberg and Suttle, 2010), and Simpler Syntax (Culicover and Jackendoff, 2006), are surface-oriented in the sense that they explicitly prohibit silent syntactic elements such as empty categories or internal structure within ellipsis sites. It is not entirely clear how the present results could be derived without positing either a silent dependency or internal structure in the ellipsis site.

One possible response is that these theories could accommodate the findings via additional stipulations. However, such a move would risk treating Experiment 4 as an isolated phenomenon, thereby missing a broader generalization. In this sense, the present results suggest that psychologically adequate theories of grammar should allow for abstract dependencies that span multiple surface-distinct constructions, even when those dependencies are not phonologically realized.

9 Experiment 5: Topicalization

Experiment 4 suggested that the comparative operator movement dependency triggers the *that*-reduction effect. Experiment 5 examined whether topicalization triggers the *that*-reduction effect, in order to test the generalizability of this effect across different subtypes of $\bar{A}$ dependencies. We specifically contrasted sentences containing topicalization, such as (13-a), with minimally different sentences containing so-called *hanging-topic left dislocation (HTLD)* (see, e.g., Villa-García 2023; Den Dikken and Surányi 2017), in order to isolate the effect of movement dependencies from more general binding or coreference relations.

At least in English, HTLD constructions do not appear to involve movement dependencies. Specifically, topicalization, but not HTLD, shows sensitivity to island constraints. Consider the following examples:

- (13) a. *This book, I heard the claim that the boy understood.

- b. This book, I heard the claim that the boy understood it.

In both cases, *this book* is interpreted as the object of *understood*, which occurs inside a complex noun phrase. However, only (13-a) is judged unacceptable. Thus, if the *that*-reduction effect is truly a reflex of successive cyclic movement, we expect it to be observed in topicalization sentences but not in HTLD sentences.

9.1 Methods

9.1.1 Participants

For the acceptability study, 42 self-reported monolingual speakers of English were recruited (35 residing in the U.S., 7 residing in the U.K.; mean age = 43; age range = [24, 69]). For the production study, 36 self-reported monolingual speakers of English were recruited (18 residing in the U.S., 18 residing in the U.K.; mean age = 46; age range = [20, 78]). Three participants were replaced because they did not follow the instructions.

9.1.2 Materials

Twenty-four sets of sentences like the following were constructed.

- (14) a. This expensive house, no one assumed (that) I could ever purchase it.
- b. This expensive house, no one assumed (that) I could ever purchase.
- c. No one assumed (that) I could ever purchase this expensive house.

Sentence (14-a) contains HTLD. Sentence (14-b) contains topicalization, a type of $\bar{A}$ movement. Sentence (14-c) is a control sentence without either construction. Critically, all sentences contained a negative polarity item (e.g., *ever*) licensed by negation in the matrix clause (e.g., *no one*).

This design was motivated by the possibility that, in the topicalization and HTLD conditions, the embedding verb (*assumed*) could be analyzed as occurring in a parenthetical structure, as in (15).

- (15) This expensive house, John assumed, I could purchase (it).

Under such a parse, the dependency between matrix negation and the embedded negative polarity item would be disrupted, since negative polarity items require licensing within the c-command domain of a negative element (at least under standard assumptions).

Each of these 24 critical sentences was paired with an unrelated distractor sentence, as in Experiments 3–5. The resulting 24 sentence pairs were intermixed with 6 filler pairs modeled on the critical items.

9.1.3 Procedure, scoring, and analysis

The same procedure, scoring, and analysis methods were used as in Experiments 3–4.

9.2 Results

9.2.1 Acceptability

Figure 10 shows mean acceptability ratings for each condition. As can be seen, both left-dislocation and topicalization sentences were marginally acceptable, scoring around 4 on average. Reflecting this, there was clear evidence that control sentences were rated higher than both types ($\hat{\beta} = 1.01$, 95% CrI = [0.79, 1.22]). There was also evidence that HTLD sentences were rated slightly higher than topicalization sentences ($\hat{\beta} = -0.19$, 95% CrI = [-0.32, -0.07]). Importantly, there was no evidence that *that* affected acceptability ($\hat{\beta} = 0.00$, 95% CrI = [-0.07, 0.06]).

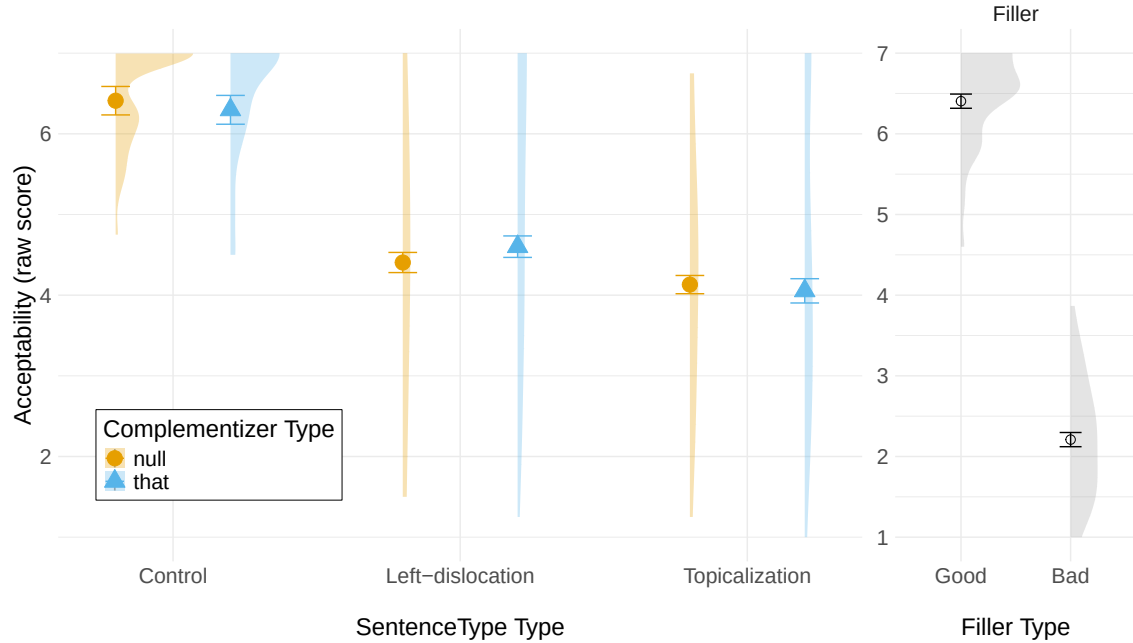


Figure 10: Mean rating of critical sentences used in Experiment 5. Error bars are within-subject standard errors.

9.2.2 That Production

767 of 864 critical trials (88.8%) met the scoring criteria (91.0% for control sentences with *that*, 96.5% for control sentences without *that*, 82.6% for HTLD sentences with *that*, 93.8% for HTLD sentences without *that*, 80.1% for topicalization sentences with *that*, and 88.2% for topicalization sentences without *that*).

Figure 11 shows the mean rate of *that* use across conditions. As can be seen, speakers were less likely to use *that* in topicalization sentences than in HTLD and control sentences ($\hat{\beta} = -0.71$, 95% CrI = [-1.37, -0.09]). There was no evidence that the rate of *that* production differed between HTLD and control sentences ($\hat{\beta} = 0.23$, 95% CrI = [-0.32, 0.78]). There was also evidence that speakers were more likely to use *that* when the original target sentence contained it ($\hat{\beta} = 1.28$, 95% CrI = [0.57, 2.00]). No other effects were observed.

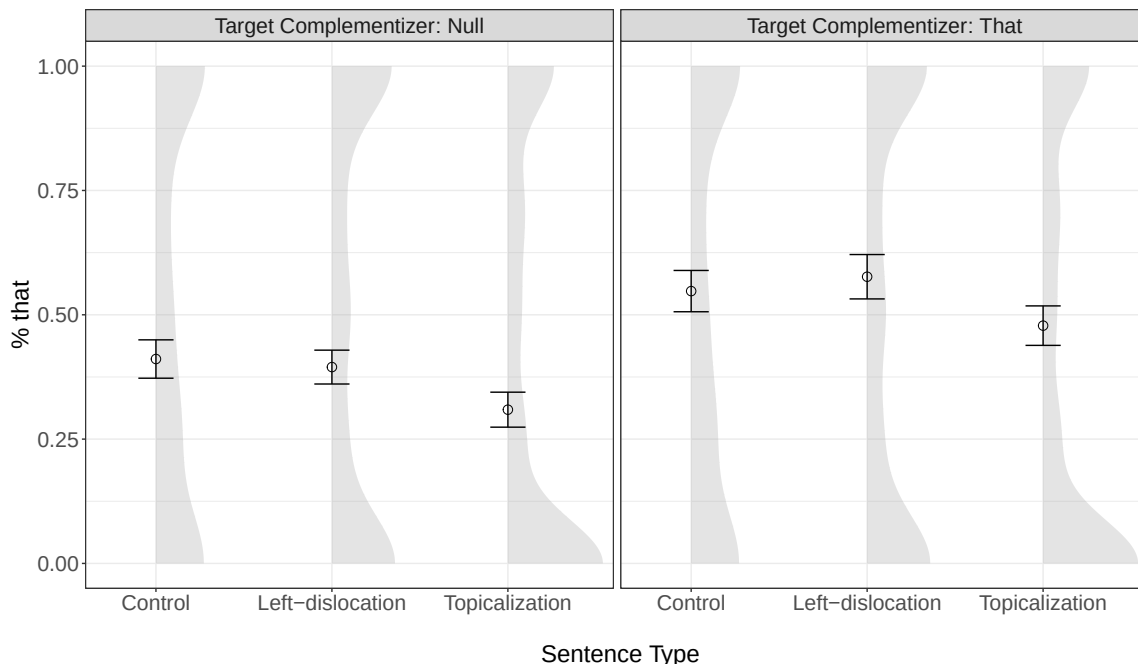


Figure 11: Mean proportion of *that* in each condition in Experiment 5. Error bars are within-subject standard errors.

These results show that topicalization triggers the *that*-reduction effect, suggesting that it is associated with $\bar{A}$ dependencies. The results further suggest that $\bar{A}$ movement dependencies, but not the binding dependencies found in HTLD, selectively trigger the *that*-reduction effect. No analogous effect was found in the acceptability judgment data.

10 Experiment 6: ATB extraction

Experiments 1–5 showed that the *that*-reduction effect is sensitive to at least three subtypes of $\bar{A}$ dependencies (four if we include Momma 2021). However, in two production studies not reported here due to space limitations, we failed to find the *that*-reduction effect in Across-the-Board (ATB) extraction sentences with VP coordination, such as the following:

(16) Which book did John read and assume (that) he understood?

This pattern was initially puzzling. However, we hypothesized that it parallels reflexes of successive cyclic movement observed in some languages, in particular Kiitharaka. We therefore treat the previously unreported studies as part of the empirical motivation for Experiments 6 and 7.

As reported by Georgi (2019, 2017), morphological reflexes of successive cyclicity in some languages respect a terminal/non-terminal distinction. That is, some reflexes are observed only when the relevant C hosts the highest copy of a moving wh-phrase, while others are observed only when it hosts intermediate copies. Georgi (2017) classifies these patterns as follows:

1. Reflex in both final and non-final clauses [Type I]
2. Reflex only in final clauses [Type II]
3. Reflex only in non-final clauses [Type III]
4. No reflex in any clauses [Type IV]

Kiitharaka pre-verbal focus marker *n-* shows a Type III pattern: it appears in clauses along the path of a $\bar{A}$ -dependency, except in the final clause where the movement terminates. The following examples are from Muriungi (2005); Abels and Muriungi (2008), as discussed in Georgi (2017):

- (17) a. I-mbi g-ug-ir-e ati John n-a-ring-ir-e?
 FOC-what 2SG-say-PERF-FV that John FOC-SM-beat-PERF-FV
 ‘What did you say that John beat?’
- b. N-uu u-ku-thugania ati John n-a-ug-ir-e Lucy
 FOC-who 2SG-PRES-think that John FOC-SM--say-PERF-FV Lucy
 n-a-ring-ir-e
 FOC-SM-beat-PERF-FV
 ‘Who do you think that John said Lucy beat?’

Importantly, Georgi (2019) shows that in ATB constructions with CP coordination in Kiitharaka (e.g., English: *What do you think that John likes and that Mary hates?*), the second conjunct does not show the reflex of movement. This suggests that movement terminates within the second conjunct in such structures. In other words, ATB constructions involve asymmetric extraction from the first conjunct (Munn, 1993; Salzmann, 2012).

If English ATB constructions likewise involve asymmetric extraction from the first conjunct, and if the *that*-reduction effect is sensitive to non-terminal wh-copies (as in Kiitharaka), then we expect no *that*-reduction effect in the second conjunct in sentences like (16). This is because, in (16), the highest C inside the second conjunct

is the C selecting the CP headed by *assume*. The wh-phrase may move higher than this position (e.g., to the edge of vP), but relative to movement triggered by C, this C hosts the terminal copy.

Thus, the absence of a *that*-reduction effect in our preliminary ATB study may reflect the fact that the effect is restricted to Cs hosting non-terminal wh-copies, combined with the assumption that ATB extraction is asymmetric from the first conjunct.

This account yields two predictions. First, the *that*-reduction effect should be observed in the first conjunct of ATB sentences, e.g., *Which book did Toma believe (that) he read and like?*

Second, the presence of the *that*-reduction effect in the second conjunct should vary with the size of the conjuncts in ATB constructions. Consider the following:

- (18) a. Which cake did Toma bake and say (that) he ate?
- b. Which cake did Toma bake and Mei say (that) she ate?
- c. Which cake did Toma bake and did Mei say (that) she ate?

We analyze (18-a) as VP coordination, (18-b) as TP coordination, and (18-c) as CP coordination (see below). Although (18-c) is sometimes judged unacceptable, we show below that (18-b) and (18-c) differ only minimally in acceptability and are both moderately acceptable for the purposes of our experimental design. If only (18-c) involves CP coordination, then we expect the *that*-reduction effect in the second conjunct only in (18-c), since only there does the relevant C fail to host the highest wh-copy within the conjunct.

In Experiment 6, we test the first prediction: that the *that*-reduction effect should be observed in the first conjunct of ATB sentences. The second prediction is tested in Experiment 7.

10.1 Methods

10.1.1 Participants

For the acceptability study, 28 self-reported monolingual English speakers were recruited (10 residing in the U.S., 18 residing in the U.K.; mean age = 51; age range = [23, 75]). For the production study, 32 self-reported monolingual English speakers were recruited via Prolific (16 residing in the U.S., 16 residing in the U.K.; mean age = 43; age range = [21, 65]). Two participants were replaced because they did not follow the instructions.

10.1.2 Materials

24 critical sentence sets like the following were constructed:

- (19) a. Which book did the boy assume (that) he understood and like?

- b. The boy assumed (that) he understood the book and liked it.
- c. Which book did the boy like and assume (that) he understood?
- d. The boy liked the book and assumed (that) he understood it.

Each sentence set was paired with an unrelated distractor sentence (e.g., *A mechanic fixed an engine to restore the vehicle.*).

For the production task, in addition to the 24 critical sentence sets, six sets of filler pairs were created. These were designed to resemble the critical items. The purpose of the filler pairs was to ensure that participants encoded both sentences in memory, including the second (distractor) sentence. In filler trials, the memory cues always corresponded to the second sentence, so participants needed to encode both sentences to perform the task successfully.

10.1.3 Procedures, scoring and analyses

The same procedures, scoring criteria, and analysis procedures as in Experiments 3–5 were used in Experiment 6.

10.2 Results and discussion

10.2.1 Acceptability

Figure 12 shows the mean ratings across conditions. Most importantly, there was no evidence that the presence or absence of *that* changed acceptability ($\hat{\beta} = -0.02$, 95% CrI = [-0.11, 0.09]). There was evidence that ATB sentences were rated lower overall than control sentences ($\hat{\beta} = 0.64$, 95% CrI = [0.48, 0.80]), and that sentences with clause embedding in the first conjunct were rated lower than sentences with clause embedding in the second conjunct ($\hat{\beta} = 0.31$, 95% CrI = [0.13, 0.50]). However, these effects were qualified by an interaction between Sentence Type and Complementizer Position ($\hat{\beta} = -0.96$, 95% CrI = [-1.24, -0.67]), suggesting that ATB sentences were rated lower than control sentences especially when clause embedding occurred in the first conjunct (see below for a potential explanation). There was no evidence for any other effects.

One notable pattern is that acceptability scores were clearly lower in ATB sentences than in control sentences when the first conjunct contained a clause embedding, as in *Which book did the boy assume (that) he understood and like?*. This degradation may reflect a type of garden-path effect. Specifically, comprehenders may initially prefer to coordinate the final VP (*like*) with the lower VP (*understood*), rather than with the higher VP (*assume (that) he understood*), in accordance with the well-known late closure principle (Frazier, 1978). This would result in a temporary parse at the point of *and* such as the following:

- (20) Which book did the boy assume he [_{ConjP} [_{VP} understood] and [_{VP} ...]]

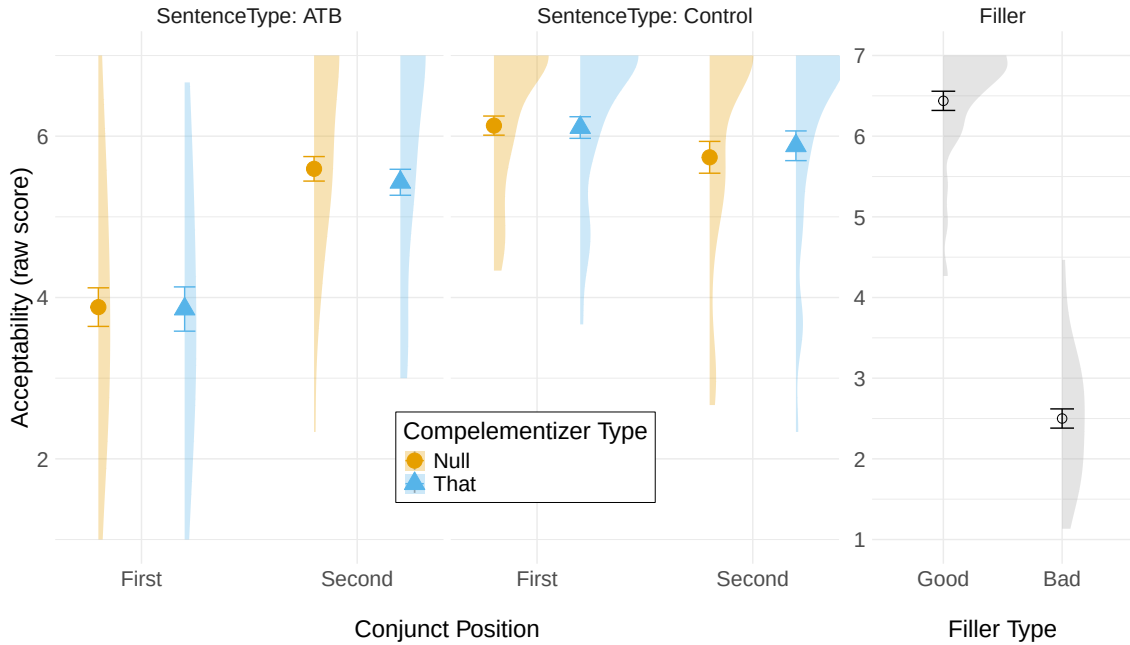


Figure 12: Mean ratings for critical (left) and filler (right) items in Experiment 6. The red and blue regions represent the distribution of subject averages. Error bars are within-subject standard errors.

The uninflected verb *like* cannot be integrated into this parse, because the structure requires an inflected verb form (e.g., *likes* or *liked*). To accommodate the uninflected verb, the following parse is required:

- (21) Which book did the boy [_{ConjP} [_{VP} assume he understood] and [_{VP} ...]

Thus, if the preferred initial parse is the first one, English speakers may experience a garden-path effect, leading to lower acceptability judgments (Warner and Glass, 1987; Ferreira and Henderson, 1991). This may explain why ATB sentences with first-conjunct extraction were rated especially low.

10.2.2 That-production

712 of 767 trials (92.7%) met the scoring criteria. The exclusion rates by condition were as follows: ATB sentences with first-conjunct clause embedding (with *that*: 85.4%, without *that*: 92.7%), control sentences with first-conjunct clause embedding (with *that*: 90.6%, without *that*: 95.8%), ATB sentences with second-conjunct clause embedding (with *that*: 91.6%, without *that*: 93.8%), and control sentences with second-conjunct clause embedding (with *that*: 94.8%, without *that*: 96.9%).

The mean rates of *that* production across conditions are shown in Figure 13. In the first-conjunct conditions, speakers were less likely to use *that* in ATB sentences

than in control sentences ($\hat{\beta} = -1.91$, 95% CrI = [-2.04, -0.89]). In contrast, in the second-conjunct conditions, there was no evidence that speakers used less *that* in ATB sentences than in control sentences ($\hat{\beta} = 0.27$, 95% CrI = [-0.46, 0.95]). There was clear evidence for a two-way interaction between Complementizer Position and Sentence Type ($\hat{\beta} = -2.18$, 95% CrI = [-3.38, -1.04]). Thus, we found the *that*-reduction effect in the first VP conjunct, but not in the second VP conjunct.

There was also evidence for an effect of Target Type, suggesting that speakers were more likely to produce *that* when it was present in the originally presented sentence.

The results show an interesting parallel to the pattern of morphological reflexes of successive cyclicity in other languages, in particular Niger-Congo languages. Georgi (2019) report that ATB sentences with CP conjuncts inside a single VP (e.g., roughly corresponding to *What do you think (that) John likes _ and (that) Mary hates _?* in the relevant languages) do not show reflexes of successive cyclic movement in the second conjunct. For example, in Kiitharaka, the pre-verbal marker *n-* appears only in the presence of a crossing $\bar{A}$ -dependency and only in non-terminal clauses.

A natural hypothesis is that the absence of the *that*-reduction effect in the second conjunct of VP-coordinated ATB sentences reflects the same structural condition that regulates the distribution of *n-* in Kiitharaka. For concreteness, assume the system proposed by Georgi (2019), in which C agrees (upward) with the closest $\bar{A}$ -moved operator, and the morphological realization of this agreement depends on whether the operator corresponds to a terminal or non-terminal copy. Under such a view, the relevant C in the second conjunct of VP-coordinated ATB sentences would systematically fail to interact with a non-terminal copy of the $\bar{A}$ -dependency, yielding a morphologically unrealized agreement in both English (which would result in the null complementizer) and Kiitharaka-like systems (which would result in the *n-*).

11 Experiment 7: ATB extraction from the second CP conjunct

In Experiment 6, we found that the *that*-reduction effect was present in the first conjunct but not in the second conjunct in ATB sentences with VP coordination. If the absence of the *that*-reduction effect is due to the relevant complementizer being the highest C inside the second conjunct, then adding a higher C inside the second conjunct should revive the effect.

Experiment 7 tested this prediction by comparing ATB sentences with TP coordination and CP coordination, as in the following:

- (22) a. Which puzzle did Emily buy and Mike assume (that) she could solve?
- b. Which puzzle did Emily buy and did Mike assume (that) she could solve?

The only difference between these sentences is the *did* in the second conjunct in (22-b). Assuming that this *did* occupies the C position, according to the standard

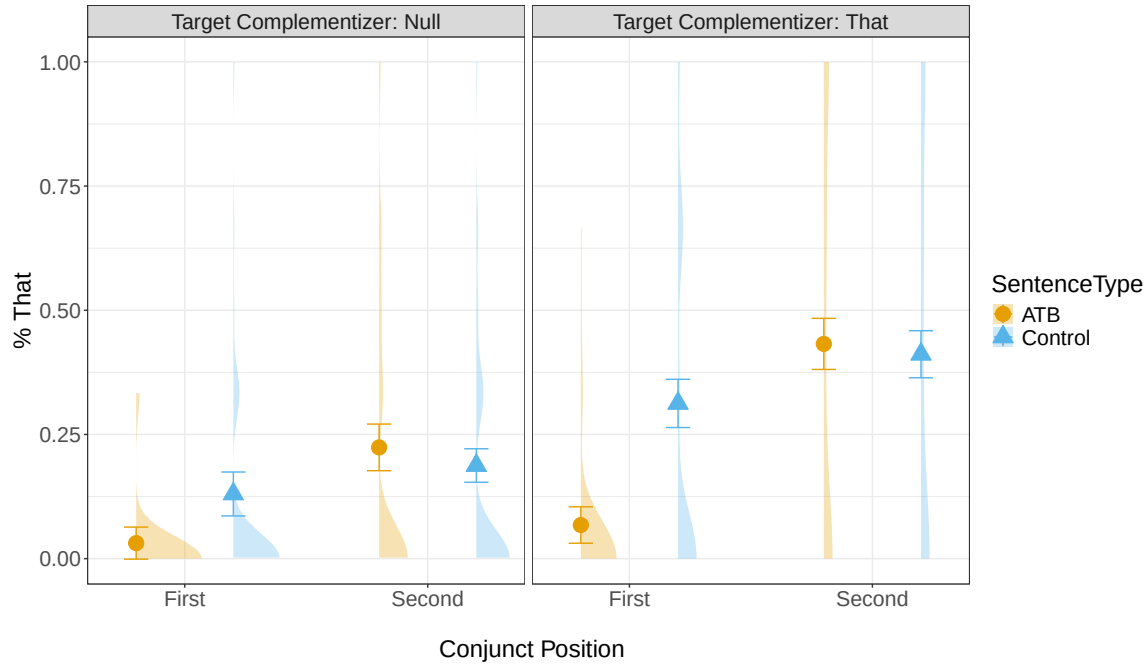


Figure 13: Mean proportion of *that* in each condition in Experiment 6. Error bars are within-subject standard errors.

assumption and assuming that in the absence of the second *did*, what is coordinated in (22-a) is TP rather than CP, (22-b) selectively has an additional CP layer in the second conjunct. Under the hypothesis that the *that*-reduction effect follows the Type III pattern in Georgi’s classification, it should only be observed in (22-b). Experiment 7 tests this prediction.

11.1 Methods

11.1.1 Participants

For the acceptability study, 28 self-reported monolingual English speakers were recruited (19 residing in the U.K., 9 residing in the U.S.; mean age = 51; age range = [23, 75]). For the production study, 64 self-reported monolingual English speakers were recruited via Prolific (29 residing in the U.K., 35 residing in the U.S.; mean age = 40; age range = [23, 73]).

11.1.2 Materials

24 sentence sets like the following were constructed by modifying the sentence sets used in Experiment 6.

- (23) a. Which puzzle did Emily buy and Mike assume (that) she could solve?

- b. Which puzzle did Emily buy and did Mike assume (that) she could solve?
- c. Emily bought the puzzle and Mike assumed (that) she could solve it.

11.1.3 Procedure, scoring, and analysis

The procedure, scoring, and analysis were the same as in Experiments 4–6, with the following exception. For the analysis of acceptability scores, we did not include VP-coordination control sentences, as they were not used in the production experiments. This resulted in a three-level factor Sentence Type: ATB sentences with TP coordination (ATB (TP)), ATB sentences with CP coordination (ATB (CP)), and control sentences with TP coordination (Control (TP)).

This three-level factor was Helmert-coded: control sentences were first compared against ATB (TP) sentences, and the mean of these two conditions was then compared against ATB (CP) sentences.

11.2 Results and discussion

11.2.1 Acceptability

Figure 14 shows the mean acceptability scores across conditions. As can be seen, ATB sentences were overall rated lower than control sentences ($\hat{\beta} = 0.47$, 95% CrI = [0.26, 0.68]). However, there was little to no evidence that the presence of *did* in both conjuncts (as in CP-coordinated ATB sentences) affected acceptability (first contrast of Sentence Type: $\hat{\beta} = -0.14$, 95% CrI = [-0.32, 0.05]). There was also no evidence that *that* affected acceptability overall (main effect of Complementizer: $\hat{\beta} = 0.01$, 95% CrI = [-0.15, 0.18]). Likewise, there was no evidence that the effect of *that* differed across sentence types (Sentence Type $\times$ Complementizer interaction: $\hat{\beta} = 0.19$, 95% CrI = [-0.18, 0.55]; $\hat{\beta} = -0.03$, 95% CrI = [-0.33, 0.26]).

Thus, ATB sentences were overall rated lower than control sentences, but the presence or absence of *that* had little to no effect, nor did the type of coordination (TP vs. CP) in ATB sentences. Importantly, all sentences, including the critical CP-coordinated ATB sentences, were rated as fairly acceptable on average, scoring around 4.5 or above on the 7-point scale (compared to clearly ungrammatical filler sentences, which received an average score below 2.5).

11.2.2 *That* production

1,319 of 1,536 trials (85.5%) met the scoring criteria (75.3% for the CP coordination sentences with *that*, 85.8% for the CP coordination sentences without *that*, 89.4% for the TP coordination sentences with *that*, 88.7% for the TP coordination sentences without *that*, 94.9% for the control sentences with *that*, and 91.0% for the control sentences without *that*).

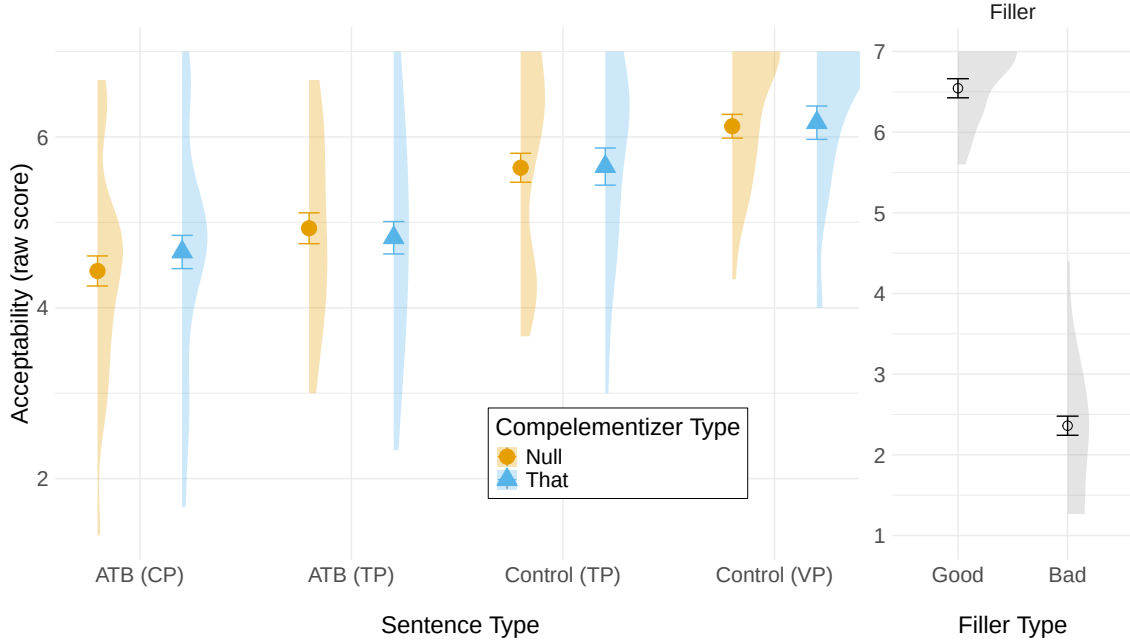


Figure 14: Mean rating of sentences used in Experiment 7. Error bars are within-subject standard errors.

The lower proportion of trials meeting the scoring criteria in the CP coordination sentences was primarily due to participants omitting the second *did* relatively often. When the second *did* was omitted, the target sentence no longer preserved the intended structure.

Figure 15 shows the mean rate of *that* production across conditions. Speakers were around 12% less likely to produce *that* in ATB sentences with CP coordination than in ATB sentences with TP coordination and control sentences (second contrast of Sentence Type: $\hat{\beta} = 0.79$, 95% CrI = [0.40, 1.18]). However, this was qualified by the interaction between the second contrast of Sentence Type and Target Complementizer ($\hat{\beta} = 1.00$, 95% CrI = [0.08, 1.91]). This suggests that the *that*-reduction effect was stronger when the original target sentence contained *that*.

There was no evidence that speakers differed in their probability of producing *that* when producing ATB sentences with TP coordination compared to control sentences (first contrast of Sentence Type: $\hat{\beta} = 0.14$, 95% CrI = [-0.05, 0.32]). Speakers were also more likely to produce *that* when the original sentence contained *that* (Target Complementizer: $\hat{\beta} = 0.91$, 95% CrI = [0.53, 1.31]), suggesting that participants retained some memory of the original sentences. There was no evidence for any other effects.

These results suggest that the *that*-reduction effect was observed inside the second CP conjunct of ATB sentences, but not in the second TP conjunct of ATB sentences, mirroring the *-n* affixation pattern in Kiitharaka. This pattern is predicted

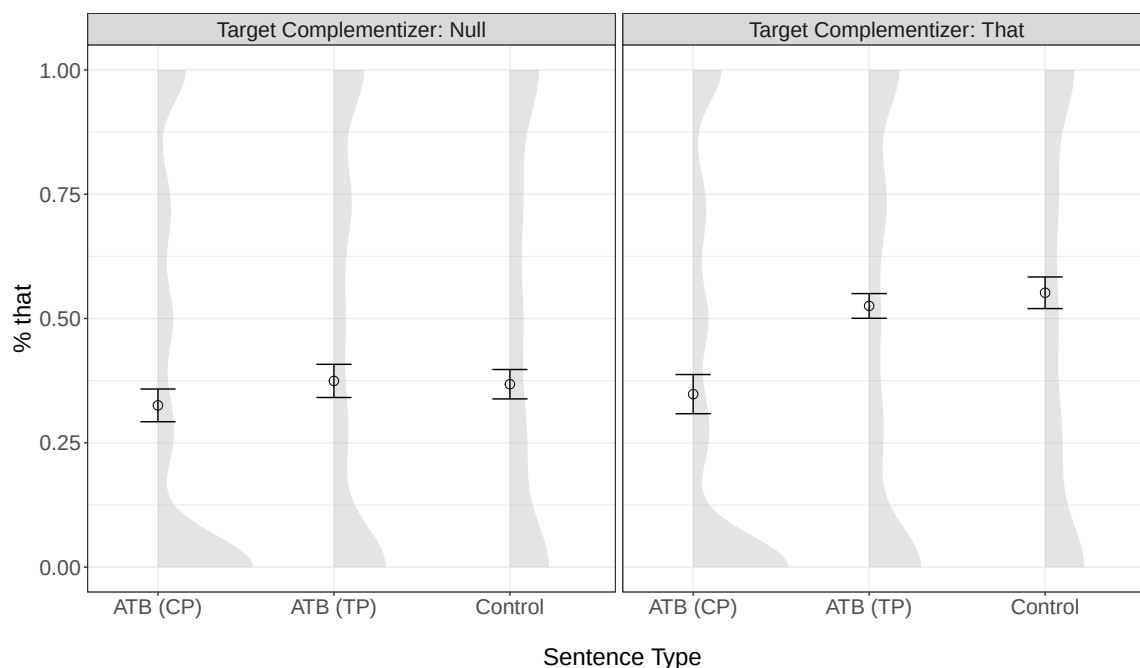


Figure 15: Mean proportion of *that* in each condition in Experiment 7. Error bars are within-subject standard errors.

if movement inside the second conjunct terminates within that conjunct, and if the *that*-reduction effect is an instance of wh-C agreement that is only realized when the closest wh-element is in a non-terminal position. We are aware of no alternative accounts that predict that the *that*-reduction effect can be selectively revived simply by adding *did* in the second clause.

A somewhat unexpected pattern is that the *that*-reduction effect in this experiment was primarily driven by trials in which the original target sentence contained *that*. We do not yet have a clear explanation for this, but one possibility is a floor effect: even when the original sentence did not contain *that*, it may be difficult in this task to reduce production rates below approximately 25%. Despite this interaction, the main finding remains clear: the *that*-reduction effect was observed selectively inside the second CP conjunct of ATB sentences.

12 General Discussion

The results of the production experiments are summarized in Table 1. Speakers were less likely to produce *that* when the relevant C lies on the hierarchical path of an $\bar{A}$ dependency. This effect was not observed in Cs that linearly, but not hierarchically, intervene in an $\bar{A}$ dependency (Experiment 1), and it holds for object wh-movement (Experiment 1), adjunct wh-movement (Experiment 2), wh-movement across more

than two clauses (Experiment 3), comparative degree operator movement (Experiment 4), and topicalization, but not closely matched HTLD involving a binding dependency (Experiment 5). Critically this effect was not observed inside the second VP or TP conjunct of ATB sentences (Experiments 6–7), but it was observed inside the first VP conjunct of ATB sentences (Experiment 6) and inside the second CP conjunct of ATB sentences (Experiment 7). These results are naturally captured if ATB constructions involve two independent $\bar{A}$ dependencies, one of which terminates inside the second conjunct (e.g., Munn 1992; see Georgi 2019 for a summary), and if English wh–C agreement is realized only when the relevant C hosts non-terminal wh-copies, as in Kiitharaka *-n* suffixation (Georgi, 2017). The key result of the current experiments, then, is that English has wh–C agreement whose realization is sensitive to the terminal/non-terminal distinction, suggesting a direct parallel with morphological reflexes of successive cyclic movement in unrelated languages such as Kiitharaka (Georgi, 2017).

Exp. #	✓: effect present, ×: effect absent Example sentences	Dependency Type
1	Who did the robber believe ✓ (that) the cashier knew? Who did the robber × (that) the cashier knew attack?	Wh
2	Where did the biologist say ✓ (that) the new species was discovered?	Wh
3	Which story did Jane imply ✓ (that) her son believed ✓ (that) she came up with?	Wh
4	The scientist wrote more papers than her co-worker assumed ✓ (that) she did.	Comparative
5	This expensive house, no one assumed ✓ (that) I could ever purchase. This expensive house, no one assumed × (that) I could ever purchase it.	Topicalization
6	Which book did the boy read and assume × (that) he understood? Which book did the boy assume ✓ (that) he understood and like?	ATB Wh
7	Which puzzle did Emily buy and Mike assume × (that) she could solve? Which puzzle did Emily buy and did Mike assume ✓ (that) she could solve?	ATB Wh

Table 1: Summary of the distribution of the *that*-reduction effect across seven production experiments.

In acceptability judgment and forced-choice tasks using the same materials as the production experiments, we found only weak evidence for the corresponding effect in 1 of 7 experiments (Experiment 4). This contrast between production and judgment is expected if wh–C agreement in English is optional, as in Chamorro. Both the presence and absence of agreement yield grammatical outputs, thereby reducing or even eliminating the sensitivity of judgment-based measures.

We discuss four broader theoretical implications. First, the results provide additional evidence that movement proceeds successive cyclically in English. If agreement relations are local, wh–C agreement implies the presence of intermediate copies of the $\bar{A}$ -moved constituent in the specifier of each CP headed by an agreeing C. The observed pattern therefore supports a successive-cyclic analysis of $\bar{A}$ movement.

Second, the results bear on the structure of ATB constructions. The selective appearance of the *that*-reduction effect in second CP conjuncts, but not in second TP conjuncts in minimally different ATB sentences, supports analyses in which extraction proceeds asymmetrically from the first conjunct. The asymmetric extraction account, together with the assumption that the *that*-reduction effect instantiates a Type III reflex in Georgi’s classification (see above), correctly predicts that the effect should be absent when the relevant C is the highest C inside the second conjunct. Symmetric analyses of ATB extraction do not straightforwardly capture this distribution.

Third, the results suggest that abstract $\bar{A}$ dependencies play an active role in shaping what people say, even when they lack phonological realization. This suggests that speakers represent potentially silent dependencies that cut across superficially distinct constructions when formulating utterances, contrary to approaches that eschew silent syntactic representations (Culicover and Jackendoff, 2006; Goldberg and Suttle, 2010).

Finally, if the *that*-reduction effect reflects wh–C agreement in English, as we propose, it may offer a new perspective on a long-standing puzzle: the *that*-trace effect (Perlmutter, 1971; Pesetsky, 2017). In both phenomena, $\bar{A}$ dependencies constrain (optionally or obligatorily) the use of the overt complementizer *that*. If the *that*-trace and *that*-reduction effects are indeed related, a unified analysis in terms of wh–C agreement may be desirable. One hypothesis we are currently developing is that wh–C agreement is optional when a non-subject wh-phrase is extracted but obligatory when a subject wh-phrase is extracted, leading to the *that*-trace effect. In the verbal agreement literature, it has been observed that in languages allowing agreement with multiple arguments, optionality is more common with lower arguments (see Akkuş et al. 2025, p. 175). It may be that both this cross-linguistic tendency and the optionality of wh–C agreement for non-subject extraction reflect a general property of agreement mechanisms.

Interestingly, the current proposal may connect the *that*-trace effect to previous observations that typically optional complementizer (number-)agreement becomes obligatory in the presence of subject extraction, as Mayr (2010) suggested for Bavarian. Under the current proposal, both the subject extraction effect on C agreement in Bavarian and the *that*-trace effect in English can be thought of as cases where subject extraction triggers obligatory realization of complementizer agreement. Thus, the current proposal may connect two seemingly disparate phenomena across languages.

Another interesting consequence of this proposal is that the well-known *that*-trace amelioration effect (Bresnan, 1977; Barss and Deprez, 1986) may also be connected to other attested patterns of C agreement. Namely, as summarized by van Koppen

(2017), in West Flemish (Ackema and Neeleman, 2004) and Bavarian (Fuß, 2008), C agrees with (non-wh) subjects, but this agreement cannot be realized when there is an intervening adjunct phrase. It is tempting to draw a parallel here: C-agreement (wh or otherwise) may not be realized in the presence of material intervening between probe and goal. This results in the lack of agreement realization in West Flemish and Bavarian, and the amelioration of the *that*-trace effect in English (note that this assumes that wh-C agreement involves downward Agree; see Georgi 2017 for discussion). This idea remains speculative, and more careful investigation is clearly needed. However, we hope to have shown that once we admit the possibility of English wh-C agreement, a new perspective on the long-standing puzzle of the *that*-trace effect becomes available, with interesting consequences.

Finally, the methodological implication of the present results is that production data can, in some cases, provide critical evidence that acceptability judgment data do not, or do so less clearly. More specifically, acceptability judgments may be systematically insensitive to optionally realized phenomena whose phonological consequences are obscured by syncretism and zero morphology, such as the optional wh-C agreement examined here.

A final note is that the *that*-reduction effect can serve as a useful diagnostic for movement dependencies, just like island effects have long served as diagnostics for movement. The *that*-reduction effect appears to track movement dependencies closely and may in addition be sensitive to the terminal/non-terminal distinction. One promising domain of application is parasitic gap constructions (see Engdahl 1983; Culicover and Postal 2001; e.g., *Which book did you read _ without buying _?*). Some theories assume parasitic gaps involve operator movement terminating inside the adjunct, while others do not (see Culicover and Postal 2001 for an overview). Preliminary work in our lab suggests that the *that*-reduction effect may be absent inside the (non-finite) adjunct clause of such sentences (e.g., *Which book did the boy read _ without assuming (that) he would understand _?*). If this pattern is confirmed, it would suggest that parasitic gaps, like the ATB constructions investigated here, involve operator movement that terminates within the adjunct. More generally, production data may sometimes provide critical evidence for syntactic theory.

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